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.NET + AI Roadmap 2026

From Scratch to AI-Ready Developer — the complete learning sequence behind our .NET Full Stack course

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.NET + AI Roadmap 2026 — From Scratch to AI-Ready Developer

1. Introduction

Modern .NET development is centered on C#, ASP.NET Core, cloud-native engineering, APIs, security, distributed systems, Azure, and AI integration. A current roadmap should not be dominated by legacy ASP.NET Web Forms or .NET Framework technologies.

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2. Complete Learning Sequence

Level 1 — Programming Fundamentals

- Algorithms
- Flowcharts
- Variables
- Data types
- Operators
- Conditions
- Loops
- Functions
- Input/output
- Debugging
- Data structures
- Time and space complexity

Practice:

- Calculator
- ATM
- Billing system
- Number guessing game
- Student marks system
- Banking application

Level 2 — .NET Platform

- .NET SDK
- .NET Runtime
- CLR
- C#
- IL
- JIT compilation
- Assemblies
- NuGet

- `dotnet` CLI
- `dotnet new`
- `dotnet build`
- `dotnet run`
- `dotnet test`
- `dotnet publish`

Level 3 — C# Fundamentals and OOP

C#:

- Variables
- Value/reference types
- Nullable types
- Type conversion
- Conditions
- Loops
- Methods
- Optional/named parameters
- `ref`, `out`, `in`, `params`

OOP:

- Classes
- Objects
- Constructors
- Fields
- Properties
- Methods
- Access modifiers
- Encapsulation
- Inheritance
- Polymorphism
- Abstraction
- Interfaces
- Composition
- Association
- Aggregation

Level 4 — Modern C#

- Delegates
- Action
- Func
- Predicate
- Lambda expressions
- Events
- Extension methods

- Records
- `init`
- `required`
- Nullable reference types
- Pattern matching
- Tuples
- Primary constructors
- Collection expressions
- File-scoped namespaces
- Global using

Level 5 — Collections, Generics and LINQ

Collections:

- Arrays
- List<T>
- Dictionary<TKey,TValue>
- HashSet<T>
- Queue<T>
- Stack<T>
- IEnumerable<T>
- ICollection<T>
- IList<T>
- IReadOnly collections

Generics:

- Generic classes
- Generic methods
- Constraints
- Covariance
- Contravariance

LINQ:

- Where
- Select
- SelectMany
- OrderBy
- ThenBy
- GroupBy
- Join
- Any
- All
- First/FirstOrDefault
- Single/SingleOrDefault
- Count

- Sum
- Average
- Min/Max
- Distinct
- Skip/Take
- Deferred execution
- IEnumerable vs IQueryable

Level 6 — Exceptions, Runtime and Async

- Exceptions
- Custom exceptions
- Exception propagation
- Stack and heap
- Garbage collection
- GC generations
- IDisposable
- `using`
- IAsyncDisposable
- Resource lifetime
- Memory leaks

Async/concurrency:

- Task
- Task<T>
- async/await
- CancellationToken
- Task.WhenAll
- Task.WhenAny
- ThreadPool
- Locks
- Concurrent collections
- Channels
- Background services
- Async streams
- IAsyncEnumerable

Level 7 — SQL and Data Access

SQL:

- SELECT, INSERT, UPDATE, DELETE
- JOIN
- GROUP BY
- HAVING
- Subqueries
- CTEs

- Window functions
- Views
- Indexes
- Transactions
- ACID
- Isolation
- Locking
- Query optimization

ADO.NET:

- Connection
- Command
- DataReader
- Parameters
- Transactions
- Connection pooling
- SQL injection prevention

Entity Framework Core:

- DbContext
- DbSet
- Entities
- Relationships
- Migrations
- LINQ queries
- Tracking
- No-tracking
- Loading strategies
- Transactions
- Concurrency

Level 8 — Dependency Injection and ASP.NET Core

- IoC
- Dependency Injection
- Service registration
- Singleton
- Scoped
- Transient
- ASP.NET Core
- Middleware
- Routing
- Configuration
- Environment
- Logging
- Options pattern

- Application lifecycle
- Minimal APIs
- Controllers

Level 9 — REST APIs and Security

REST:

- HTTP
- GET, POST, PUT, PATCH, DELETE
- Status codes
- Headers
- JSON
- DTOs
- Validation
- Pagination
- Filtering
- Sorting
- Versioning

Security:

- Authentication
- Authorization
- ASP.NET Core Identity
- Password hashing
- Roles
- Claims
- Policies
- JWT
- OAuth 2.0
- OpenID Connect
- Refresh tokens
- CORS
- CSRF
- Security headers

Level 10 — Full Stack, Testing and Dev Tools

Frontend option:

- HTML
- CSS
- JavaScript
- TypeScript
- React
- Components
- Props
- State

- Hooks
- Forms
- Routing
- API integration
- Authentication

Testing:

- xUnit
- NUnit concepts
- Assertions
- Mocking
- Integration tests
- Testcontainers
- Postman
- REST testing
- Playwright

Tools:

- Git
- GitHub
- Pull requests
- Code reviews
- GitHub Actions

Level 11 — Production Engineering

Docker:

- Dockerfile
- Images
- Containers
- Volumes
- Networks
- Docker Compose
- Environment variables
- Health checks

Redis:

- Caching
- Cache-aside
- TTL
- Distributed caching
- Sessions
- Rate limiting
- Distributed locks

Messaging:

- RabbitMQ
- Kafka
- Producers
- Consumers
- Topics
- Partitions
- Consumer groups
- Offsets
- Ordering
- Idempotency

gRPC: - Protocol Buffers - Unary calls - Streaming - Service contracts

Level 12 — Microservices, Azure and DevOps

Microservices:

- Service boundaries
- API Gateway
- Service discovery
- Inter-service communication
- REST/gRPC
- Resilience
- Circuit breakers
- Distributed transactions
- Saga pattern
- Event-driven architecture
- Modular monolith as an alternative

Azure:

- App Service
- Container Apps
- Functions
- Storage
- Azure SQL
- PostgreSQL
- Key Vault
- Monitor
- Application Insights
- Service Bus
- Container Registry
- Entra ID

CI/CD:

- GitHub Actions
- Build
- Test

- Docker
- Deploy
- Secrets
- Environment management

Observability:

- Structured logging
- Metrics
- Tracing
- Health checks
- OpenTelemetry
- Application Insights
- Prometheus/Grafana concepts

Level 13 — AI for .NET Developers

LLM fundamentals:

- Tokens
- Context windows
- Prompts
- System instructions
- Structured outputs
- Streaming
- Function/tool calling
- Model selection

Modern .NET AI:

- Microsoft.Extensions.AI
- AI abstractions
- Semantic Kernel
- Azure OpenAI
- Microsoft AI ecosystem

Build AI features in ASP.NET Core:

- Chat
- Streaming
- Conversation history
- Structured output
- Tool calling
- Authentication
- Logging
- Rate limiting

Level 14 — Embeddings, RAG and Agents

Embeddings:

- Text-to-vector representation
- Similarity
- Cosine similarity
- Semantic search

Vector search:

- PostgreSQL + pgvector
- Azure AI Search
- Other vector databases

RAG:

- Document ingestion
- Parsing
- Chunking
- Embeddings
- Retrieval
- Reranking
- Context construction
- Grounded answers
- Citations
- Evaluation

Tool calling:

- Database tools
- Search
- CRM
- Inventory
- Email
- Calendar
- Internal APIs

Agents:

- Planning
- Tool use
- Memory
- State
- Multi-step workflows
- Human-in-the-loop

Level 15 — AI Security and Evaluation

Security:

- Prompt injection
- Data leakage
- Excessive permissions
- Tool authorization

- Input/output validation
- Rate limiting
- Audit logging
- Human approval

Evaluation:

- Accuracy
- Relevance
- Groundedness
- Hallucination
- Retrieval quality
- Latency
- Token usage
- Cost

Level 16 — Architecture

- SOLID
- DRY
- KISS
- Layered architecture
- Clean Architecture
- Hexagonal architecture
- Modular monolith
- Microservices
- Event-driven architecture
- CQRS concepts
- Saga
- Distributed systems
- Scalability
- Fault tolerance
- Load balancing

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3. Recommended Project Progression

1. Banking Console Application
2. Employee Management System
3. Student Management with LINQ
4. Inventory System with SQL
5. E-commerce backend with EF Core
6. Secure ASP.NET Core API
7. React + ASP.NET Core full-stack application
8. Redis-enabled API
9. Kafka order-processing platform
10. Microservices e-commerce system

11. Azure-deployed application
12. AI chat application
13. Company knowledge assistant
14. AI business assistant with tool calling
15. Enterprise AI-powered .NET platform

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4. Recommended 2026 Stack

C# + modern .NET + ASP.NET Core + EF Core + PostgreSQL/SQL Server + React + TypeScript + Redis + Kafka + Docker + Azure + GitHub Actions + Microsoft.Extensions.AI + Semantic Kernel + AI APIs + RAG + AI Agents

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5. What Not to Prioritize Initially

Do not make these the center of a new learning path:

- ASP.NET Web Forms
- Old .NET Framework-first development
- Legacy MVC patterns without current relevance
- Memorizing every framework API
- Learning several frontend frameworks simultaneously

Learn legacy technologies when a specific maintenance project or job requires them.

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6. Career Outcome

Possible roles:

- C# Developer
- .NET Developer
- ASP.NET Core Developer
- Backend Developer
- .NET Full Stack Developer
- Cloud Developer
- Azure Developer
- Microservices Developer
- AI-enabled .NET Developer
- Enterprise Application Developer

7. Core Principle

AI should accelerate .NET development, not replace C# and software engineering fundamentals.

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This roadmap is the learning sequence our .NET Full Stack Development course follows. The live, always-current version — with week-by-week detail, projects, batch timings and fees — is on the course page linked above.